International Journal of Geo-Information

MDPI

Article

What Daily Nighttime Lights Can and Cannot Show About Power Outage and Recovery: Winter Storm Uri in Harris County, Texas

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Abstract

Daily nighttime light (NTL) images are widely used to map power outages and recovery, but clouds and quality screening leave gaps. We examined what NASA Black Marble data show about the February 2021 Winter Storm Uri outage in Harris County, Texas. When only high-quality retrievals were used, 16 February was the only usable night during the outage, and the next were 19 to 21 February. On 16 February, 5112 of 13,846 developed pixels (37%) were below half their pre-storm brightness, and 92% of those with an observed return were lit by 21 February. Return dates between these nights are unobservable. Recovery dates from a detector-based workflow depend on smoothing, persistence and aggregation choices; in a controlled experiment with known recovery dates and flag-0 observations only, an unsmoothed rule requiring three consecutive calendar nights has a median detection delay of 12 to 15 days. Compared with 20 ordinary clear nights, 16 February dimming was far more extensive and clustered. However, the estimated average total impact of the Social Vulnerability Index was 10.6 percentage points of reference radiance (95% interval −32.0 to 49.8), within the range estimated on ten ordinary pre-storm nights (−31.9 to 26.8). Single-night community comparisons need an ordinary-night baseline.

Keywords: Winter Storm Uri; nighttime lights; Black Marble; power outage; recovery detection; observation availability; placebo test; social vulnerability; spatial regression

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Academic Editors: Wolfgang Kainz and Godwin Yeboah

Received: 27 August 2026
Revised: 20 October 2026
Accepted: 27 October 2026
Published: 29 October 2026

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1. Introduction

Winter Storm Uri brought several days of sub-freezing temperatures to Texas in February 2021. As generation failed, the Electric Reliability Council of Texas (ERCOT) ordered rotating outages from the early hours of 15 February, and more than 4.5 million customers lost power [1,2]. Harris County, which contains Houston, was among the hardest hit. In a University of Houston survey, 91% of county residents said they lost power at some point during the storm [3]. Knowing where power was lost, and for how long, matters for emergency response and for judging whether some communities bore more of the burden than others [4–7].

Utility outage records are rarely available below the county or ZIP code level [8–10], so researchers often use daily nighttime light (NTL) data instead [11]. The Day/Night Band of the Visible Infrared Imaging Radiometer Suite (VIIRS) records nighttime light every night, with finer resolution and better calibration than earlier sensors [12,13]. NASA's Black Marble products turn these records into daily radiance at 15 arc-seconds (about 500 m),

corrected for moonlight, the atmosphere and viewing geometry [14,15]. Daily NTL data have been used to map outages after hurricanes, earthquakes and winter storms [16–20], to follow recovery over days to months [18,21,22], and to compare outage or recovery between population groups [16,18,22]. Such comparisons draw on research asking whether socially vulnerable communities, often identified with the CDC/ATSDR Social Vulnerability Index (SVI) [23], experience more or longer outages. The answers vary with the measure, the spatial scale and the event [8–10,24–27], and composite indices such as the SVI carry their own measurement uncertainty [28,29].

Daily observations are irregular. Clouds, snow, quality screening and the fixed overpass time leave many nights without a usable value for a given pixel. Even when nothing changes on the ground, radiance varies from night to night with viewing geometry, snow and atmospheric conditions [16,22,30]. Studies deal with this by comparing radiance with a pre-event baseline [18,20–22], by smoothing or aggregating the daily series [18,20] and, when they need a recovery date, by requiring the return to baseline to last for several consecutive days [21]. Two questions are rarely asked. How precisely can such rules date recovery with the observations actually available during an event? And do community differences measured on a single night stand out from the differences that appear on ordinary nights?

Three prior studies are particularly relevant to our analysis. Xu et al. [16] mapped Uri outages across Texas from the Black Marble image of 16 February, the night with the most cloud-free coverage during the outage. They adjusted radiance for viewing angle and snow, chose the outage threshold that best matched county outage-tracking data ($R^2 = 0.42$) and found a weak tract-level correlation between outage share and the Hispanic population share ($r = 0.083$). Zheng et al. [22] modeled daily radiance after ten hurricanes with Bayesian change detection. Data gaps covered about 31% of days in their record; they tested detected changes against pre-hurricane variation and showed that aggregating to tracts or counties hides much of the variation in recovery over months. Best et al. [10] used ZIP code outage records at 15 min intervals with spatial Durbin models and reported direct and indirect effects of income on restoration times after Hurricane Isaac. The outage caused by Uri lasted days, not months, and the gaps in the satellite record were as long as the outage itself. For such an event, it is not known how precisely satellite data can date the return of light.

We address these questions for Harris County. Our objectives are to (1) identify the nights with usable observations and find out what they show about where light was lost and when it returned; (2) test how precisely detector-based workflows can date recovery, using a controlled experiment with known recovery dates and a check of how tract averages depend on implementation and aggregation choices; and (3) compare 16 February with ordinary clear nights, including differences by social vulnerability and population density, and test how these estimates change with the reference period, land cover mask, spatial weights and model.

Section 2 describes the data and methods. Section 3 presents the results, Section 4 interprets them, Section 5 discusses the study's limitations, and Section 6 concludes.

2. Materials and Methods

2.1. Study Area and Event Timeline

Figure 1 shows the location of Harris County, Texas, and the spatial distribution of the census tracts included in this study. Located in southeastern Texas, the county covers approximately 4604 km^2. At the time of Winter Storm Uri in February 2021, it had an estimated population of approximately 4.74 million, making it the third most populous county in the United States [31].

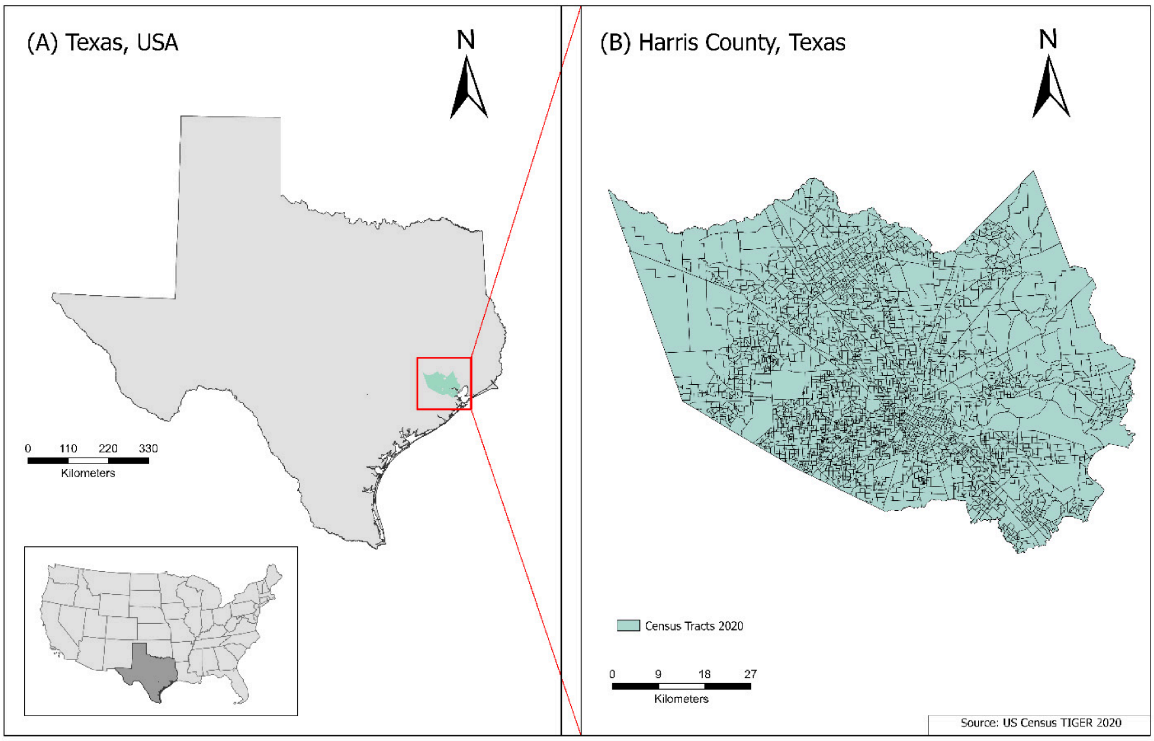


Figure 1. Location and study area of Harris County, Texas. (**A**) Location of Harris County within Texas and Texas within the United States. (**B**) 2020 census tract boundaries in Harris County used for the analysis.

Harris County has a dense urban core around downtown Houston, with suburban, rural and industrial land toward its edges. We compare communities at the level of 2020 census tracts and measure dimming at the level of satellite pixels. The study set contains 1070 of the county's 1115 tracts (96.0%), after removing 6 tracts with no resident population and 39 tracts with fewer than five developed pixels. Each analysis uses a subset of these tracts, stated in Sections 2.3–2.7.

ERCOT ordered rotating outages in the early hours of 15 February 2021 [2]. In the Environment for Analysis of Geo-Located Energy Information (EAGLE-I) record [32], the share of Harris County customers without power rose sharply on the evening of 15 February, stayed mostly between 14% and 20% until the afternoon of 17 February and fell below 1% early on 18 February (Section 3.1). EAGLE-I is compiled from utility reports and covered roughly 82% to 94% of Texas customers in 2021 [32]. Its values therefore show the county timeline, not exact customer counts.

2.2. Data

Table 1 lists the data and Figure 2 summarizes the workflow. Satellite data were processed in Google Earth Engine and all statistical analyses were done in R 4.3.3.

Table 1. Data used in this study.

Dataset	Variables Used	Resolution	Period
NASA Black Marble VNP46A2, Collection 2 [14,15]	DNB_BRDF_Corrected_NTL; Mandatory_Quality_Flag	15 arc-seconds (about 500 m), daily	10 January to 27 March 2021
NASA Black Marble VNP46A1 [14]	UTC_Time (overpass time)	15 arc-seconds, daily	15 to 21 February 2021

Table 1. *Cont.*

Dataset	Variables Used	Resolution	Period
MODIS MOD10A1 and MYD10A1, Version 6.1 [33,34]	NDSI_Snow_Cover	500 m, daily	15 and 16 February 2021
EAGLE-I outage records [32]	Customers out; modeled customers per county	County, 15 min	8 February to 27 March 2021
CDC/ATSDR SVI 2020 [23]	RPL_THEMES; E_TOTPOP; AREA_SQMI	Census tract	2016 to 2020 ACS
NLCD 2019	Land cover classes 21 to 24	30 m	2019
TIGER/Line 2020; Census cartographic boundary file 2020	Tract boundaries (TIGER/Line in Earth Engine; cartographic boundaries for pixel assignment and spatial weights)	Census tract	2020

ACS = American Community Survey; CDC/ATSDR = Centers for Disease Control and Prevention/Agency for Toxic Substances and Disease Registry; EAGLE-I = Environment for Analysis of Geo-Located Energy Information; NDSI = Normalized Difference Snow Index; NLCD = National Land Cover Database; SVI = Social Vulnerability Index.

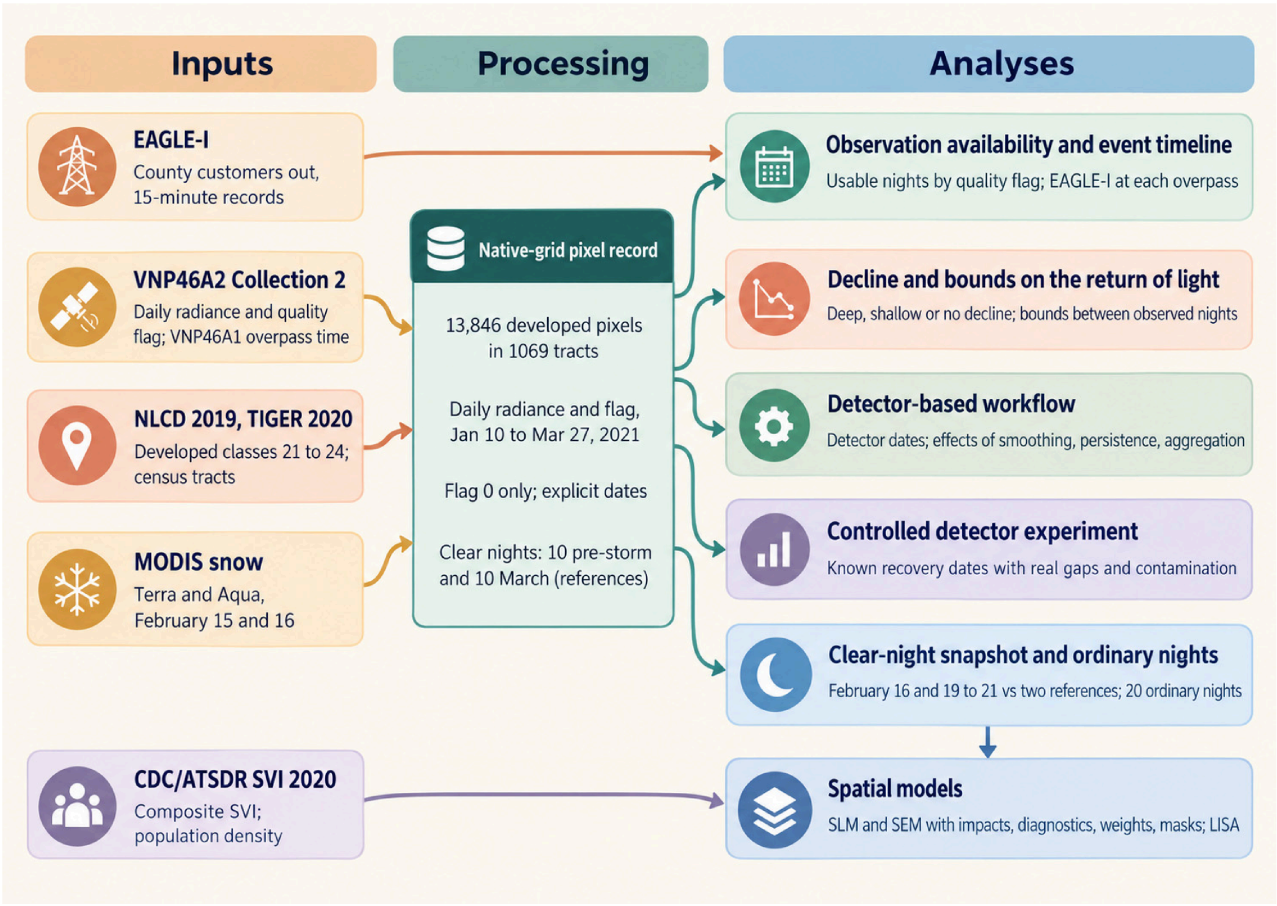


Figure 2. Workflow of the analysis. Arrows show which inputs feed each analysis. The satellite analyses use the native-grid pixel record (flag-0 observations, with the median of clear nights as references); the EAGLE-I outage records are used only for the event timeline, and the SVI only in the spatial models. The detector-based workflow (Section 2.5) is tested against the observations.

2.2.1. Nighttime Lights, Quality Flags and Dates

We used the DNB_BRDF_Corrected_NTL band of the NASA Black Marble VNP46A2 product, Collection 2 [14,15]. This band is corrected for atmospheric, lunar and viewing-

geometry effects; the separate gap-filled band was not used. Each pixel-night carries a Mandatory_Quality_Flag. In Collection 2, flag 0 marks a high-quality retrieval, flag 1 a poor-quality retrieval (an outlier, possible cloud contamination or other problems), flags 2 to 5 other poor-quality conditions and 255 no retrieval [15]. Between 10 January and 27 March 2021, our study area contains only flags 0, 1, 5 and 255. We used only flag-0 observations.

Each image was given the local calendar date of its overpass. On the key nights, the UTC_Time layer of VNP46A1 gives overpass times between 01:26 and 02:54 local time (02:48 on 16 February and 01:52 on 19 February). All date ranges in this paper include both end dates. Because filterDate in Earth Engine excludes the end date, we set it one day later.

2.2.2. Social Vulnerability and Population Density

We measured social vulnerability with the composite percentile ranking (RPL_THEMES) of the CDC/ATSDR SVI 2020 [23], which is based on 2016 to 2020 American Community Survey five-year estimates. Values run from 0 (least vulnerable) to 1 (most vulnerable), so a one-unit change in SVI spans the whole range of the index. In the spatial lag models of Section 2.8, the size of an SVI association is summarized by impacts, not by the SVI coefficient alone. Population density is total population (E_TOTPOP) divided by land area (AREA_SQMI), log-transformed and standardized within each analysis set.

2.2.3. Land Cover, Snow and Outage Records

The main pixel set contains the pixels whose center falls on NLCD 2019 class 22, 23 or 24 (developed, low to high intensity). For the mask sensitivity tests we also computed each pixel's share of 30 m cells in classes 22 to 24 and in classes 21 to 24. Class 21 (developed open space) includes large-lot residential land. Class 24 includes commercial and industrial land as well as dense housing, so none of these masks removes all non-residential lighting.

Snow status on 15 and 16 February came from the MODIS Terra (MOD10A1) and Aqua (MYD10A1) daily snow products [33,34]. A pixel counted as snow when NDSI_Snow_Cover was at least 10 and as unobserved when MODIS made no snow decision. We read the EAGLE-I county record at each satellite overpass time and use it only to check the timing of the event (Section 3.1). It cannot validate tract-level dimming: it counts customers for the whole county, whereas the satellite measure counts pixels within tracts, and snow can brighten pixels whether or not power is on.

2.3. Pixel Record and Clear Nights

All analyses used one Earth Engine export. It holds the daily record, from 10 January to 27 March 2021 (77 images), of every native pixel (15 arc-second grid) in Harris County that contains any NLCD class 21 to 24. For each pixel and night, the radiance and the quality flag are provided, and for 14 to 19 February the MODIS snow status. Each pixel was assigned to the tract that contains its center, using the 2020 Census cartographic boundary polygons of the study tracts. Of the 22,049 exported pixels, 21,341 lie inside the study tracts. The main set is the 13,846 pixels whose center lies on class 22, 23 or 24, in 1069 tracts. The other pixels were used only in the mask sensitivity tests.

We call a night clear when at least 90% of the main set had a flag-0 retrieval. This gives ten pre-storm clear nights (12, 14, 15, 16, 18, 28 and 31 January, and 1, 3 and 7 February) and ten March clear nights (3, 4, 7, 8, 13, 18, 19, 20, 21 and 26 March). These two groups are the reference pools.

2.4. Decline During the Outage and Bounds on the Return of Light

For each pixel, the relative radiance on night t is

$$r_{p,t} = \frac{L_{p,t}}{R_p} \quad (1)$$

where $L_{p,t}$ is the flag-0 radiance of pixel p on night t and the reference R_p is the median of $L_{p,t}$ over the pre-storm clear nights with a flag-0 retrieval for that pixel (at least five were required). A ratio of 1 means the pixel was as bright as on ordinary pre-storm clear nights, 0.5 means half as bright and values above 1 mean brighter. We classified each pixel by its flag-0 ratios in the storm window, 15 to 18 February. A pixel declined if its lowest ratio was below 0.9, showed no decline if all its ratios were at least 0.9, and had insufficient observation if it had no flag-0 retrieval in the window. Declines were split into deep (lowest ratio below 0.5) and shallow (0.5 to 0.9). To judge these cut-offs, we computed the same ratios for each pre-storm clear night against the median of the other nine.

For declined pixels, we bounded the return of light with observations instead of dating it with a detector:

$$u_p = min\{t > d_p : r_{p,t} \geq 0.9\}, l_p = max\{t < u_p : r_{p,t} < 0.9\} \quad (2)$$

where d_p is the last flag-0 night between 15 and 18 February with a ratio below 0.9, and only flag-0 nights count. The light returned after the lower bound l_p and on or before the upper bound u_p; the nights in between were not observed. Pixels with no observed return by 27 March are reported as unresolved. Elapsed time is counted from 15 February.

2.5. Detector-Based Recovery Dates

A detector turns the daily series into a recovery date: a pixel counts as recovered on the first day its radiance returns to the pre-event baseline, or to a set share of it, and stays there for several days [21]. We first implemented such a workflow for this event. Its baseline was the median radiance from 10 January to 8 February, and a pixel recovered on the first of three consecutive days with smoothed radiance at or above 90% of that baseline. The tract recovery time (days to recovery, DTR) was the mean over the tract's developed pixels. The code and outputs of this workflow are in Supplementary File S1.

Several implementation details of this workflow turned out to matter, and they are easy to miss (Table 2). It kept flag-1 retrievals, which are poor quality in Collection 2. Its smoothing window, written as filterDate(D − 1, D + 1), excludes the end date in Earth Engine and is therefore a two-day trailing median, which also gives values to nights without a retrieval. It numbered days from 10 February (day 1), five days before the outages began, did not require a decline before recovery, and averaged pixels that never met the rule into tract means as 47 days. Finally, it aggregated pixels to tracts on a 500 m grid in geographic coordinates (EPSG:4326) with area weights, instead of on the native grid of the product. The rest of the analysis avoids each of these choices.

We tested this workflow in two ways. First, to see how much the tract pattern depends on aggregation, we re-implemented its rule in R on the native-grid pixel record, checked that it gives the same pixel recovery days as Earth Engine, and changed one aggregation step at a time: the reduction grid, the area weighting of boundary pixels and the assignment of pixels to tracts by their center. As a sensitivity diagnostic, we also split each tract's pixels at random into two halves 200 times and correlated the half-tract means across tracts. The Spearman–Brown formula converts this correlation into the reliability of a full-tract mean if the pixels of a tract are treated as interchangeable samples of one tract value. Where recovery truly differs within a tract, low agreement between halves also reflects that real variation, so the diagnostic shows how much a tract mean depends on which pixels contribute, not

measurement error alone. Second, we compared its dates, and those of simpler detectors, with the observation-based bounds of Section 2.4. The simpler detectors used no smoothing or a centered three-day median of available observations, and persistence of one or three nights. Persistence was counted either as consecutive calendar nights, where a missing night breaks the run, or as consecutive available observations, where missing nights are skipped. For each detector we report the share of dates that fall on nights with no observation.

Table 2. Choices in the detector-based workflow and in the rest of the analysis.

Choice	Detector-Based Workflow	Rest of the Analysis
Quality flags	Flags 0 and 1	Flag 0 only (flag 1 is poor quality in Collection 2)
Smoothing	Two-day trailing median (D − 1, D); nights without a retrieval took the previous value	None for timing; smoothing tested as a variant
Dates	46 images, 10 February to 27 March; day 1 = 10 February	10 January to 27 March; elapsed time counted from 15 February
Decline before recovery	Not required	Required; deep (<0.5) and shallow (0.5 to 0.9) declines separated
Persistence	Three consecutive smoothed images	Not used for timing; 1, 2, 3 and 5 nights tested
Pixels that never meet the rule	Averaged as 47 days	Reported as unresolved
Spatial handling	500 m grid in EPSG:4326; area-weighted tract means	Native pixels assigned to tracts by their center
Outcome	Tract mean of pixel recovery days (DTR)	Bounds on the return of light for declined pixels; clear-night relative radiance for tracts

2.6. Controlled Detector Experiment

To find out how precisely recovery can be dated with the observations actually available, we applied detectors to simulated series with known recovery dates. Each of the 13,842 pixels with at least five pre-storm clear nights was given the trajectory $L_{p,t} = R_p\, e_{p,t}\, f(t)$. Here, $f(t)$ is 0.2 from 15 February until the night before the true recovery date and 1 afterwards, and $e_{p,t}$ is lognormal noise with the pixel's own standard deviation of $\log(L/R)$ on pre-storm clear nights (median 0.29, capped at 1). Noise is drawn independently for each pixel and night, without temporal or spatial correlation. In each scenario all pixels share one true recovery date: 16, 17, 18, 19, 20, 22 or 25 February, or 1 or 8 March. Because each run is one random realization, we repeated the whole experiment with ten random seeds (42 and 1 to 9), drawing new noise and contamination values each time, and checked the results with seeds 10 to 49.

We then imposed each pixel's real quality flags in three settings, to separate the effect of gaps from that of contamination. In setting A, only flag-0 nights are observed, with clean values. In setting B, flag-0 and flag-1 nights are observed with clean values, which reproduces the gaps of the detector-based workflow. Setting C is the same as B, except that on each flag-1 night the clean value is replaced by a ratio drawn from real flag-1 observations on non-storm dates (10 January to 9 February and 8 to 27 March). This adds contamination without letting any storm-period radiance into the simulation.

The detectors combined three kinds of smoothing (none, the two-day trailing median of the detector-based workflow or a centered three-day median), four persistence lengths (1, 2, 3 or 5 nights) and the two counting rules, searching from February 15. The detector-based workflow itself, which searches from February 10, was applied to settings B and C. Detection error is the detected date minus the true date. Pixels with no detection by 27 March are counted as undetected. Median errors and the shares of exact and early dates are calculated

among detected pixels, and undetected pixels are reported separately. For each detector we summarize these measures across the five true dates from 16 to 20 February and the ten seeds.

2.7. Clear-Night Snapshot, Reference Nights and Placebo Comparison

We measured community differences on clear nights instead of through a recovery duration. For each pixel we computed two ratios to each reference (the pre-storm and the March pool): one for 16 February radiance and one for the median radiance of 19 to 21 February. The snapshot set contains the pixels with flag 0 on 16 February, on at least two of 19 to 21 February and on at least five nights in each pool. Tract values are the mean of the pixel ratios multiplied by 100, for tracts with at least five such pixels. We call this percent of reference: 100 means the tract was as bright as on ordinary clear nights, and 60 means it had 40% less light. Differences between tracts are therefore in percentage points. This gives 13,582 pixels in 922 tracts. We also modeled the ratio of the March to the pre-storm reference, to show how much the choice of reference alone changes the outcome. To check sensitivity to small denominators, we counted pixels with a reference below 5 nW cm^{-2} sr^{-1}, recomputed tract means without them and compared the mean of ratios with the ratio of tract means.

Snow can brighten nighttime radiance by reflecting artificial light. For each pixel we combined the four MODIS Terra and Aqua images of 15 and 16 February: the pixel was observed if any image made a snow decision, and had snow if any observation showed snow. Each tract was then classed as unknown when fewer than 90% of its pixels were observed, as snow when at least 10% of observed pixels had snow, and otherwise as low snow. Snow status uses the same 13,582 pixels and 922 tracts as the snapshot set. Because MODIS observes in daylight, it cannot detect snow present only at night.

To judge whether 16 February differed from ordinary nights, we computed the same measures for each of the 20 reference nights, each tested against the median of the other nine nights in its pool. All nights use one common set: the 10,340 pixels with flag 0 on 16 February and on all 20 reference nights, in the 824 tracts with at least five such pixels, with one spatial weights matrix. For each night we report the shares of pixels below 0.5 and above 1.5 of the reference, Moran's I of the tract means and the spatial lag model of Section 2.8. Ordinary nights contain real spatial differences in lighting and observation conditions, so the spread of their estimates shows background variation. It is not a false-positive rate.

2.8. Spatial Models and Diagnostics

We modeled tract outcomes with spatial lag models (Equation (3)) and, as a check, with spatial error models (Equation (4)) [35]. The predictors were SVI and log population density, and the weights W were row-standardized queen contiguity weights [36,37]:

$$y = \rho W y + X\beta + \varepsilon \tag{3}$$

$$y = X\beta + u,\ u = \lambda W u + \varepsilon \tag{4}$$

where y is the vector of tract outcomes, X holds the intercept, SVI and log density, ρ is the spatial autoregressive parameter, λ the spatial error parameter and $\varepsilon \sim \mathrm{N}(0, \sigma^2 I)$. In a spatial lag model, a change in one tract passes on to its neighbors, so we converted the coefficients to direct, indirect and total impacts [38]:

$$S_k(W) = (I - \rho W)^{-1}\beta_k,\ Direct_k = n^{-1} tr(S_k(W)),\ Total_k = n^{-1}\iota' S_k(W)\iota \tag{5}$$

where β_k is the coefficient of predictor k, n the number of tracts and ι a vector of ones. Averaged over tracts, the direct impact is the response of a tract's own outcome to a one-unit change in its own predictor, the total impact combines this own-tract response with

the spillover responses through neighboring tracts, and the indirect impact is the total minus the direct impact. Impact point estimates are evaluated at the fitted parameters, and their 95% intervals are the 2.5th and 97.5th percentiles of 1000 simulated draws. The SVI coefficient itself is reported as a model parameter, and the size of an association is interpreted through the impacts. We checked residual spatial dependence with Moran's I of the model residuals and computed robust Lagrange multiplier tests on ordinary least squares residuals.

As sensitivity tests, we refitted the models with rook and six-nearest-neighbor weights; with the land cover masks of Section 2.2.3 and a mask that drops main-set pixels at least 50% in class 24 (high intensity); and with two extra controls, development intensity (the tract mean share of each pixel in classes 22 to 24) and the log distance from each tract to downtown Houston (City Hall).

Local indicators of spatial association (LISA) [39] were computed for 16 February relative radiance with 999 permutations. We report clusters at $p < 0.05$, after false-discovery-rate control [40] and after Bonferroni correction, and use them only to describe where dimming was concentrated. A bivariate local Moran's I of SVI against neighbors' 16 February relative radiance (499 permutations, false-discovery-rate control) was computed on the common set for 16 February and for each ordinary clear night. Analyses used the R packages sf (version 1.0.15), spdep (version 1.3.1) and spatialreg (version 1.3.1) [36,37]. Figures were drawn with the R packages ggplot2 (version 3.4.4) and patchwork (version 1.2.0). The original workflow diagram for Figure 2 was created with matplotlib (version 3.11.2) in Python, and its visual presentation was subsequently redesigned using ChatGPT's image-generation feature, as described in Section 2.9.

2.9. Code Development and Verification

The analysis and figure code was written by the authors with the assistance of AI tools (see Acknowledgments) and reviewed by the authors before use. Two kinds of checks were made. M.J.A. verified the complete R workflow: the scripts in Supplementary File S1 were run in order on a clean copy of the supplied exports and reproduced every output table used in the paper, the R implementation of the detector-based workflow gave the same recovery day as the Earth Engine output for all 13,846 pixels (Section 3.3), and every number reported in the text and tables was checked against these outputs. Separately, C.X. recalculated the decline categories, the bounds on the return of light, the snapshot and common sets, the snow classes and the main spatial-model point estimates from the exported data with an independent Python (version 3.14.5) implementation, and obtained the same values.

ChatGPT's image-generation feature was used to redesign the visual presentation of the authors' original workflow diagram in Figure 2. The authors checked the text, numerical values and workflow connections against the original diagram.

3. Results

3.1. Observation Availability During the Event

Figure 3 places the satellite observations on the outage timeline. No night from 9 to 15 February had any flag-0 retrievals, and neither did 17 or 18 February (18 February had flag-1 retrievals for 20.0% of pixels). 16 February is the only usable night during the outage, with flag 0 for 98.2% of pixels. The next usable nights are 19 February (99.7%), 20 (99.8%) and 21 (87.5%). At the overpass, EAGLE-I recorded 15.3% of modeled customers out on 16 February, 2.2% on 18 February (at the nominal overpass time) and 0.05% on 19 February. The county median relative radiance was 0.83 on 16 February, and 0.92, 1.09 and 1.27 on 19, 20 and 21 February. The satellite record can therefore show which pixels were dark

early on February 16 and which were lit again by 19 to 21 February, but not what happened in between.

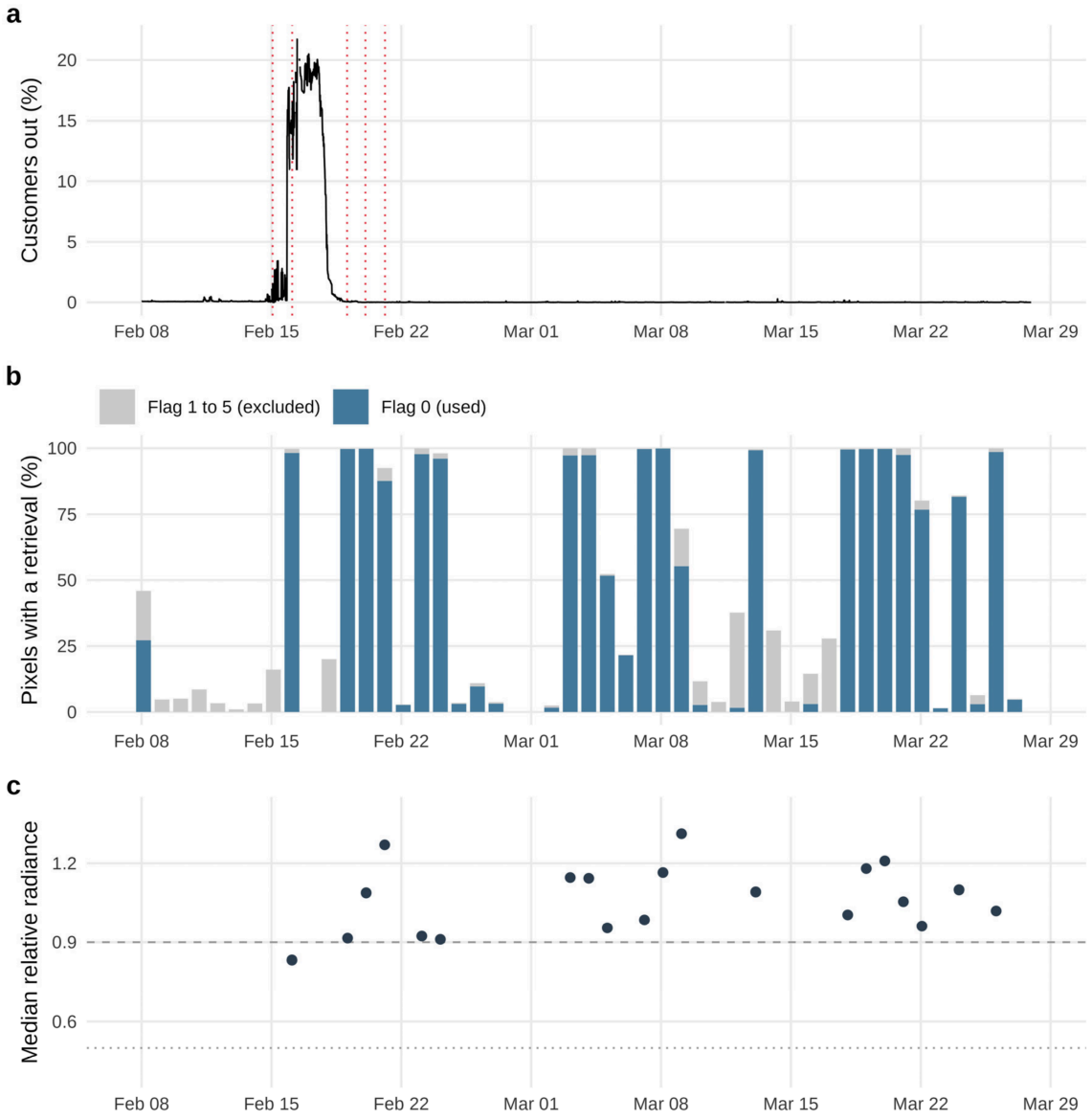


Figure 3. Event timeline. (**a**) Customers without power in Harris County (EAGLE-I), as a percentage of modeled customers; red dotted lines mark satellite overpasses with a measured time. Readings during feed dropouts (values below 1% with values above 10% within three hours on both sides, and single readings between such gaps; 17 of 4595) are removed. (**b**) Share of developed pixels with a retrieval each night, by quality flag. (**c**) Median relative radiance (radiance divided by the pre-storm reference) on nights with flag 0 for at least half of the pixels; dashed and dotted lines mark 0.9 and 0.5.

3.2. Dimming and the Return of Light

Of the 13,846 pixels, 7137 (51.5%) declined during the storm window, 6463 (46.7%) showed no decline and 242 (1.7%) had no usable observation; four had no valid reference (Table 3). Whether a decline stands out depends on ordinary variation. On ordinary pre-storm clear nights, each tested against the other nine, 19.4% to 50.8% of main-set pixels fall below 0.9 of their reference, but only 1.7% to 7.0% fall below 0.5. Shallow declines (2025 pixels) are thus within ordinary variation, while deep declines (5112 pixels) are not. The no-decline group cannot be read as uninterrupted service, because only one outage night was observed.

Table 3. Storm-period decline categories and observation-based bounds on the return of light (pixels).

Category	Pixels	% of Pixels	Return Observed by 21 February	Return Observed After 21 February	No Return by 27 March
Declined, deep (minimum ratio < 0.5)	5112	36.9	4723	386	3
Declined, shallow (0.5 to 0.9)	2025	14.6	1889	135	1
No decline observed	6463	46.7			
Insufficient observation	242	1.7			
No valid reference	4	<0.1			

Bounds for deep declines with an observed return (n = 5109): after 16 February and on or before 19 February, 2556 (50.0%); after 19 February and on or before 20 February, 1730 (33.9%); after 20 February and on or before 21 February, 431 (8.4%); other bounds, 392 (7.7%). Ratios are relative to the pre-storm clear-night reference; flag-0 observations only.

Within the storm window, every declined pixel's dark observation was on 16 February, the only usable night in that window. For most deep declines, light was back on one of the next clear nights (Table 3). Of the 5109 deep declines with an observed return, light returned after 16 February and by 19 February for 50.0%, by 20 February for a further 33.9% that were still dark on 19 February, and by 21 February for 8.4% that were still dark on 20 February. Another 386 first showed light after 21 February, and three had not by 27 March. Some of these late returns are probably noise near the threshold: on ordinary clear nights the middle half of pixels in each category lies between about 0.6 and 1.7 of the reference, and single pixels vary more (Figure 4). For most declined pixels, then, the observations show that light was absent early on 16 February and present again by 19 to 21 February, but not when within that interval it returned.

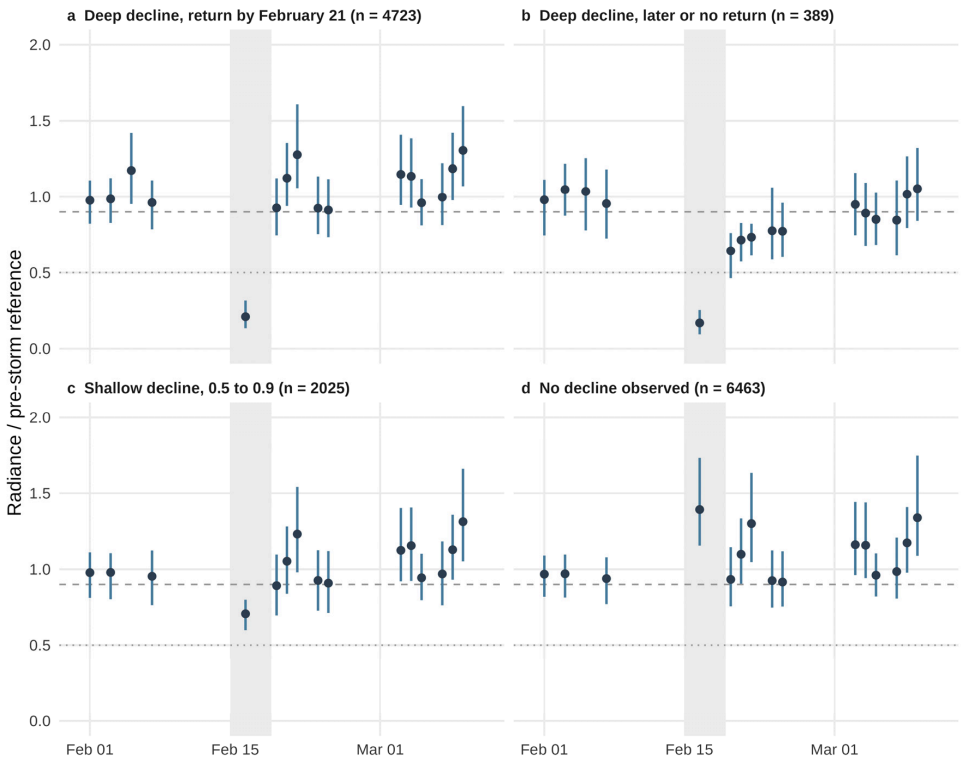


Figure 4. Daily relative radiance (flag-0 radiance divided by the pre-storm clear-night reference; 1 = as bright as usual) by decline category, 1 February to 10 March 2021: (**a**) deep decline, return by 21 February; (**b**) deep decline, later or no return; (**c**) shallow decline, 0.5 to 0.9; (**d**) no decline observed. Points show the median and bars the interquartile range of all pixels in the category, on nights when at least half of them had a flag-0 observation; n is the number of pixels. The grey band marks 15 to 18 February; dashed and dotted lines mark 0.9 and 0.5 of the reference.

3.3. How Precisely Detectors Can Date Recovery

Detectors applied to the same pixels give dates that look more precise than the observations allow. A centered three-day median put 60% of its dates on nights with no observation. A single-observation rule without smoothing placed recovery on the first observed return (median 20 February). Three consecutive calendar nights moved the median to 3 March and left 20.5% of declined pixels undetected.

The controlled experiment shows how much of this comes from the rules and the gaps (Table 4, Figure 5). With true recovery between 16 and 20 February, and across the ten seeds, a single flag-0 observation without smoothing had a median detection delay of 0 to 2 days. Longer persistence added delay and missed pixels: the median delay was 2 to 6 days for three available observations (5% to 8% undetected), 5 to 11 days for five available observations (41% to 46%) and 12 to 15 days for three consecutive calendar nights (28% to 37%). The rule of three available observations varied most between seeds, because a detected date can fall only on an observed night and small changes in the noise move the median from one observed night to the next. A centered three-day median had median errors of −1 to +1 days but dated up to 65% of detected pixels before the true date. The median delay of the detector-based workflow was 1 to 3 days in setting B and 1 to 4 days in setting C. Both settings include the flag-1 nights, so their difference shows the effect of contamination on those nights under this observation mask, at most one day here, and not the overall effect of including flag 1. When the true return fell in a cloudy spell (25 February), most rules had median delays of 6 to 10 days in every seed. For this event, the spacing of usable nights sets the attainable precision.

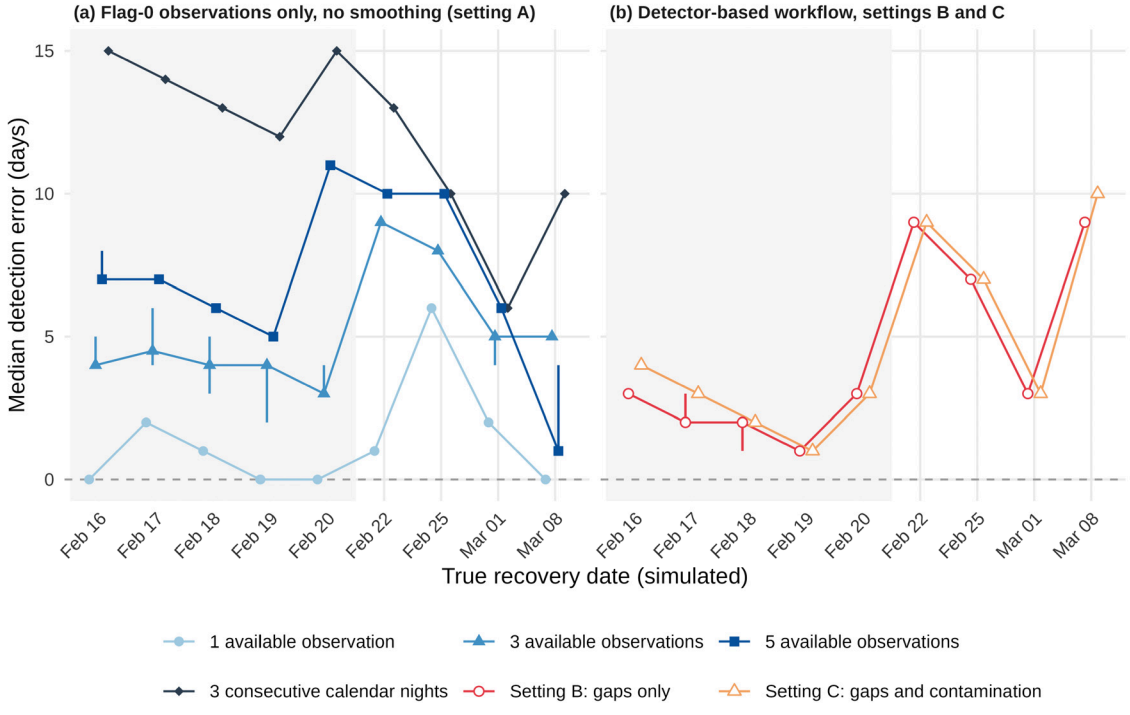


Figure 5. Median detection error by true recovery date in the controlled experiment: (**a**) rules without smoothing applied to flag-0 observations only (setting A); (**b**) the detector-based workflow with flag-0 and flag-1 nights observed, with clean values (setting B) or with contamination on flag-1 nights (setting C). Points show the median across ten random seeds and bars the range across seeds (no bar where all seeds agree). The shaded dates, 16 to 20 February, are those summarized in Table 4. The dashed line marks zero error.

Tract averages add a second problem. The R and Earth Engine versions of the detector-based workflow gave the same recovery day for all 13,846 pixels, but the tract pattern depended on aggregation (Table 5). Moving the same pixel values from the 500 m EPSG:4326 grid to the native grid changed tract values substantially ($r = 0.85$; mean absolute difference 1.03 days), and assigning pixels by their center, without area weights, changed them further ($r = 0.94$; 0.69 days). Overall, the two versions have almost the same county mean (14.51 and 14.55 days) but correlate at only $r = 0.78$ across tracts. Even with the same rule and the same pixel values, the Earth Engine and R tract means differed slightly ($r = 0.98$; mean absolute difference 0.07 days), because the two assigned some pixels near tract boundaries to different tracts: Earth Engine used the TIGER/Line polygons and R the generalized cartographic boundary polygons. Membership differed by one or two pixels in 110 of the 1069 tracts, and 17 tract means differed by more than one day. The largest difference, 16.5 days, was in a tract with only one or two pixels, where a single pixel just outside the generalized boundary changed the mean.

Table 4. Controlled detector experiment: detection error with true recovery dates of 16 to 20 February, across ten random seeds.

Detector	Setting	Median Error (Days)	Exact Date, Best Case (%)	Undetected (%)
Single observation, no smoothing	A	0 to +2	66	0
Three available observations, no smoothing	A	+2 to +6	32	5 to 8
Three calendar nights, no smoothing	A	+12 to +15	36	28 to 37
Five available observations, no smoothing	A	+5 to +11	26	41 to 46
Five calendar nights, no smoothing	A	+26 to +30	3	87 to 88
Centered three-day median, three calendar nights	A	−1 to +1 (up to 65% early)	55	0
Detector-based workflow (from February 10)	B	+1 to +3	38	1
Detector-based workflow (from February 10)	C	+1 to +4	38	1 to 2

Setting A: flag-0 nights observed; B: flag-0 and flag-1 nights observed, clean values; C: as B with contamination on flag-1 nights drawn from non-storm flag-1 observations. Median error and the exact-date share are calculated among detected pixels; undetected pixels are reported separately. Ranges are across the five true dates and ten random seeds (42 and 1 to 9), and the exact-date share is the highest value in any of these runs. Repeating the experiment with seeds 10 to 49 gave the same ranges of median error for the configurations in this table. Full results for all detectors, true dates and seeds are in Supplementary File S1.

The day numbering and the detection rule both make these recovery times look long, but in different ways. The workflow numbered days from 10 February (day 1), five days before the outages began on 15 February. This numbering inflates the reported duration but does not change the detected calendar date. 19 February, for example, is day 10 on this numbering. Moving the origin to 15 February accounts for five days (19 February would be day 5), and counting elapsed time instead of numbering days from 1 accounts for one more: 19 February is four days after 15 February. A county mean of about 14.5 on this numbering corresponds to a detected date around 23 or 24 February, about 8.5 days after 15 February, when EAGLE-I recorded almost no customers out. The detected dates themselves fall several days after the first clear nights after the outage, by which time most light had already returned (Section 3.2); this delay comes from the smoothing and the persistence rule (Table 4).

Tract means were also sensitive to which pixels contributed. Recovery days varied widely among pixels in the same tract (median within-tract standard deviation 6.2 days), and the means of two random halves of a tract's pixels correlated at only $r = 0.21$ across tracts, which the Spearman–Brown formula converts to 0.35 for a full-tract mean. Part of this within-tract variation may be real, because recovery can differ between neighborhoods

of the same tract, so this diagnostic does not separate noise from heterogeneity. It does show that the tract pattern from this workflow depends strongly on which pixels are included.

Table 5. Tract recovery time (DTR) from the detector-based workflow under different aggregation steps.

Comparison	Tracts	r	Mean DTR (Days)	Mean Absolute Difference (Days)
500 m EPSG:4326 grid vs. native grid, both area-weighted	1070	0.85	14.51 vs. 14.52	1.03
Native grid: area-weighted vs. pixel center in tract	1069	0.94	14.53 vs. 14.56	0.69
500 m EPSG:4326 grid, area-weighted vs. native grid, pixel center	1069	0.78	14.51 vs. 14.55	1.34

Earth Engine (pixel center, native grid) and R gave the same recovery day for all 13,846 pixels and nearly the same tract values (r = 0.98; mean absolute difference 0.07 days; 17 tracts differ by more than one day, maximum 16.5 days), because pixels near tract boundaries were assigned with different polygons (TIGER/Line in Earth Engine, cartographic boundaries in R). Split-half diagnostic for tract means of pixel recovery days: median within-tract standard deviation 6.2 days; correlation of half-tract means r = 0.21 (median of 200 random splits); Spearman–Brown value for a full-tract mean 0.35.

These sensitivities matter for comparisons between communities. In the persistence sensitivity run of the detector-based workflow (images to March 23, 90% threshold), the spatial lag SVI coefficient for tract recovery time was −0.54 days ($p = 0.047$) with three-day persistence and +1.04 days ($p = 0.039$) with five-day persistence. Requiring five consecutive days instead of three adds a median of 11 days per pixel (mean 13.7). Of the pixels with a three-day run, 43.4% had the run broken on its fourth or fifth day by a smoothed value below the threshold (35.6% on a night with an actual observation), 16.9% by a missing night, and 39.7% continued.

Observation availability barely differs between SVI quartiles (46.1% to 46.8% of nights usable under the mask of the detector-based workflow; Table S1). Ordinary clear-night variability, however, is slightly higher in more vulnerable tracts. SVI correlates at r = 0.15 with the share of pre-storm clear-night observations below 0.9 of the reference, and at r = 0.16 with the delay added by the five-day rule. These results are consistent with runs breaking more often in more vulnerable tracts, which would be enough to reverse the sign of a small coefficient. They do not fully establish the cause of the reversal, but they show that the sign depends on how the detector handles noisy observations and cannot be read as a difference in recovery.

3.4. 16 February Compared with Ordinary Clear Nights

On the common set (10,340 pixels in 824 tracts), 16 February stands out from ordinary nights in the amount and clustering of dimming, but not in brightening (Table 6, Figure 6). With the pre-storm reference, 35.1% of pixels were below half their reference on 16 February, against 1.2% to 5.9% on the ten ordinary nights. With the March reference the figures were 37.0% and 1.6% to 5.9%. Moran's I of tract means was 0.57 and 0.58 on 16 February, against at most 0.38 and 0.24 on ordinary nights. The share of pixels brighter than 1.5 times the reference (19.3% and 13.6%) was within the ordinary range (7.3% to 20.4% and 6.2% to 16.3%). Brightening, which snow could cause, was therefore no greater on 16 February than on ordinary nights.

The association with SVI behaves differently (Figure 7). The SVI coefficient varied widely from one ordinary night to the next: from −23.7 to 23.4 with the pre-storm reference (standard deviation 15.7) and from −14.5 to 10.4 with the March reference (standard deviation 9.4), and its 95% confidence interval excluded zero on 19 of the 20 nights. On 16 February it was 2.0 (95% CI −5.9 to 10.0) and −0.6 (−7.6 to 6.4), near the middle of the ordinary-night range. The size of the association is summarized by the average total impact

of SVI, which combines own-tract and spillover responses to a one-unit change in SVI. On 16 February it was 10.6 percentage points of reference radiance (95% interval −32.0 to 49.8) with the pre-storm reference and −3.3 (−42.9 to 33.6) with the March reference, against ranges of −31.9 to 26.8 and −16.9 to 13.2 on ordinary nights (Table 6). The intervals are wide because spatial dependence was strong ($\rho = 0.81$). In other words, the association between SVI and brightness on 16 February was no stronger than on a typical clear night. The ordinary-night estimates are not false positives, because those nights contain real spatial differences in lighting and observation conditions. They show that a single-night SVI coefficient from these data varies far more from night to night than its own confidence interval suggests, and that the 16 February coefficient cannot be told apart from this background.

Table 6. 16 February compared with ordinary clear nights on a common set of 10,340 pixels in 824 tracts.

Measure	Ordinary Nights, Pre-Storm Pool ($n = 10$)	16 February, Pre-Storm Reference	Ordinary Nights, March Pool ($n = 10$)	16 February, March Reference
Pixels below 0.5 of reference (%)	1.2 to 5.9	35.1	1.6 to 5.9	37.0
Pixels above 1.5 of reference (%)	7.3 to 20.4	19.3	6.2 to 16.3	13.6
Moran's I of tract mean ratio	0.03 to 0.38	0.57	0.00 to 0.24	0.58
SVI coefficient	−23.7 to 23.4 (SD 15.7)	2.0 (−5.9 to 10.0)	−14.5 to 10.4 (SD 9.4)	−0.6 (−7.6 to 6.4)
SVI average total impact (percentage points)	−31.9 to 26.8	10.6 (−32.0 to 49.8)	−16.9 to 13.2	−3.3 (−42.9 to 33.6)
Density coefficient (per SD)	−2.5 to 0.8	0.5 (−1.9 to 2.8)	−2.6 to 1.1	0.7 (−1.5 to 2.8)
Nights with SVI coefficient 95% CI excluding zero	10 of 10	No	9 of 10	No
Bivariate LISA tracts (FDR < 0.05)	0 on every night	78	0 on every night	87

Ordinary nights are each tested against the median of the other nine nights in the same pool. Coefficients are from spatial lag models of tract mean relative radiance (×100) on SVI and log population density, with 95% confidence intervals. The average total impact combines own-tract and spillover responses to a one-unit change in SVI, in percentage points of reference radiance, and is evaluated at the fitted parameters with a simulated 95% interval. FDR = false discovery rate; SD = standard deviation.

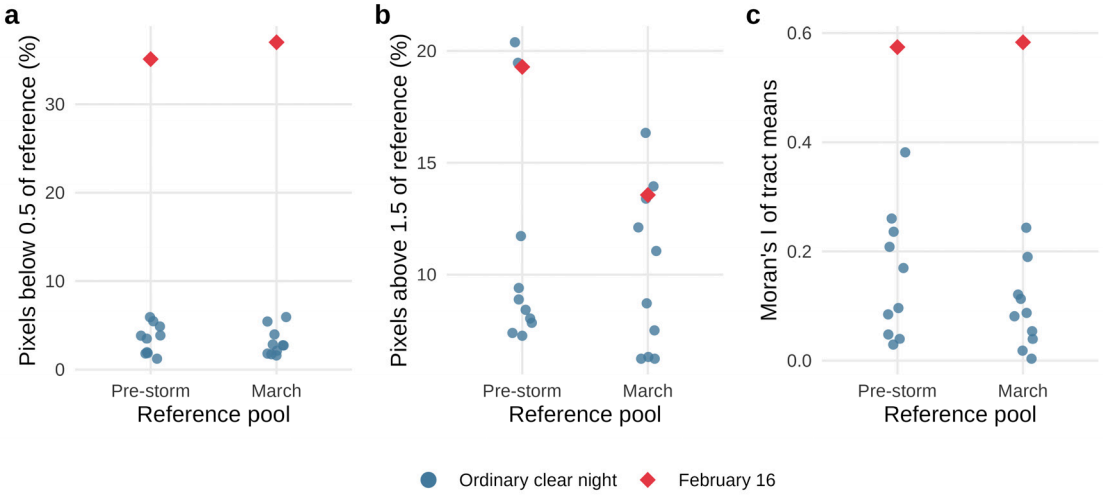


Figure 6. 16 February (red) against ordinary clear nights (blue) for each reference pool, measured on the same pixels and tracts: (**a**) pixels below 0.5 of the reference; (**b**) pixels above 1.5 of the reference; (**c**) Moran's I of tract mean relative radiance.

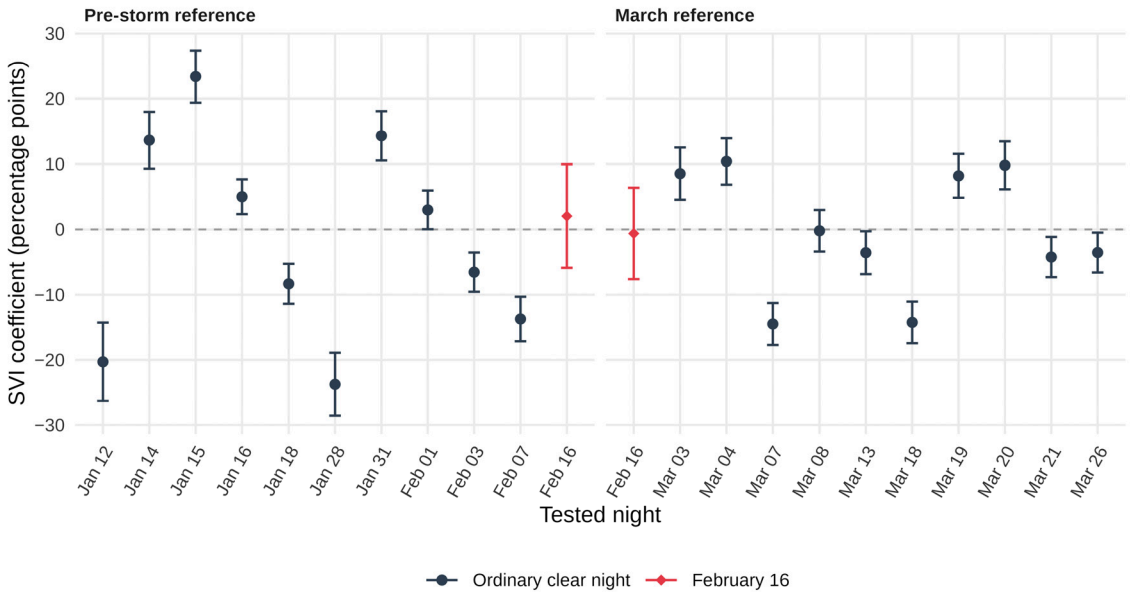


Figure 7. SVI coefficient of the spatial lag model (with log population density) and its 95% confidence interval on each tested night. The dashed line marks zero. 16 February in red; ordinary clear nights in dark blue. The coefficient is a model parameter; the corresponding average total impacts are summarized in Table 6.

The density coefficient on 16 February, 0.5 (−1.9 to 2.8) and 0.7 (−1.5 to 2.8), was also within the ordinary-night range. In spatial error models it was negative, −2.2 (−5.2 to 0.9) and −1.8 (−4.4 to 0.9), and again within the ordinary-night ranges (−2.6 to 0.9 and −2.6 to 1.0).

Two further checks, for small denominators and for snow, did not change this picture. Pixels with a reference below 5 nW cm^{-2} sr^{-1} made up 0.3% to 0.6% of the common set. Dropping them changed tract means by 0.03 to 0.33 percentage points, and the mean of ratios correlated with the ratio of means at 0.89 to 0.98. Of the 922 snapshot tracts, 495 had low snow, 201 had snow and 226 were unknown because MODIS observed fewer than 90% of their pixels (Table S2). In the 696 tracts with adequate observation, the snow share had a small and imprecise association with 16 February relative radiance (0.06 percentage points per percentage point of snow, 95% CI −0.04 to 0.16), and with snow share in the model the SVI coefficient was 4.7 (−4.7 to 14.1). In the 495 low-snow tracts, the SVI coefficient was 4.9 (−7.3 to 17.1) with the pre-storm reference and 2.1 (−8.5 to 12.8) with the March reference, similar to all tracts.

3.5. Community Differences Across Analytical Choices

Table 7 reports the main models for the 922 snapshot tracts. 16 February relative radiance averaged 94.7% of the pre-storm reference and 86.2% of the March reference, with large differences between tracts (standard deviations of 59.0 and 53.1 points). The SVI coefficient was 3.9 (95% CI −4.3 to 12.2) with the pre-storm reference and 1.7 (−5.9 to 9.3) with the March reference. Spatial dependence was strong ($\rho = 0.82$ and 0.80), so the average total impacts of SVI, which add spillover responses to the own-tract response, were larger and much less precise than the coefficients: 21.4 percentage points of reference radiance (95% interval −22.7 to 67.8) and 8.5 (−28.1 to 46.3). These intervals are compatible with no association, but also with large positive or negative associations. The density coefficients were −0.95 (−3.44 to 1.55) and −0.73 (−3.02 to 1.56) per standard deviation. Residual Moran's I was close to zero in every model (−0.010 to 0.007, $p \geq 0.34$). Robust Lagrange

multiplier tests on ordinary least squares residuals did not clearly favor either model for 16 February (robust lag test $p = 0.09$ and 0.07; robust error test $p = 0.25$ and 0.40).

Table 7. Spatial lag models of tract relative radiance (percent of reference) on the snapshot set (922 tracts).

Outcome (Night/Reference)	SVI Coefficient	SVI Direct Impact	SVI Indirect Impact	SVI Total Impact	Density Coefficient	ρ	Residual Moran's I (p)
16 February/pre-storm	3.9 (−4.3 to 12.2)	4.9 (−5.1 to 15.4)	16.5 (−18.6 to 52.5)	21.4 (−22.7 to 67.8)	−0.95 (−3.44 to 1.55)	0.82	0.002 (0.43)
16 February/March	1.7 (−5.9 to 9.3)	2.1 (−6.9 to 10.9)	6.4 (−21.8 to 35.7)	8.5 (−28.1 to 46.3)	−0.73 (−3.02 to 1.56)	0.80	−0.010 (0.67)
19 to 21 February/pre-storm	11.5 (8.6 to 14.4)	11.6 (8.5 to 14.6)	3.0 (1.6 to 4.7)	14.6 (10.9 to 18.2)	−0.96 (−1.79 to −0.13)	0.21	−0.005 (0.57)
19 to 21 February/March	3.5 (1.3 to 5.8)	3.5 (1.4 to 5.8)	0.9 (0.3 to 1.7)	4.4 (1.8 to 7.2)	−0.01 (−0.68 to 0.67)	0.21	0.004 (0.40)
March reference/ pre-storm reference	7.7 (5.7 to 9.7)	7.7 (5.7 to 9.7)	1.1 (0.3 to 2.0)	8.8 (6.7 to 11.1)	−1.38 (−1.96 to −0.79)	0.13	0.007 (0.34)

Coefficients with 95% confidence intervals. Impacts are average responses to a one-unit change in SVI, in percentage points of reference radiance (direct: own-tract response; indirect: spillover response; total: both), evaluated at the fitted parameters with simulated 95% intervals (1000 draws). Density coefficient per standard deviation of log population density. Queen contiguity weights, row-standardized.

The SVI coefficients for 16 February changed little across specifications. With rook or six-nearest-neighbor weights, and with spatial error models, the coefficient (pre-storm reference) ranged from 3.9 to 8.7, and all intervals included zero (Table S3). Across the five land cover masks it ranged from 3.9 to 4.8, again with all intervals including zero (Table S4). This includes the two masks that add class 21 (4.6, −3.8 to 13.0; and 4.3, −3.9 to 12.4) and the mask that drops pixels at least 50% in class 24 (4.8, −5.3 to 15.0; 815 tracts). With development intensity and distance to downtown added, it was 5.0 (−3.5 to 13.4) with the pre-storm reference and 2.6 (−5.1 to 10.3) with the March reference.

The density coefficients depended on the model. On the snapshot set they were negative in every specification. In the spatial lag models their intervals included zero, except with the broadest land cover mask (any class 21 to 24: −2.8, −5.2 to −0.3). In the spatial error models they were −3.2 to −4.0 per standard deviation, with intervals excluding zero (for example −4.0, −7.1 to −0.8, with queen weights and the pre-storm reference), and these models fitted slightly better by AIC (9510.5 against 9512.7 with queen weights). Development intensity had positive coefficients (3.4, −0.4 to 7.1; and 3.8, 0.4 to 7.2), so more developed tracts tended to dim less relative to their reference, although the interval included zero with the pre-storm reference.

For the nights after the outage, the choice of reference mattered. Relative radiance on 19 to 21 February rose with SVI against the pre-storm reference (coefficient 11.5, 95% CI 8.6 to 14.4; average total impact 14.6 percentage points of reference radiance, 10.9 to 18.2), but much less against the March reference (coefficient 3.5, 1.3 to 5.8; average total impact 4.4, 1.8 to 7.2). The gap arises because the March reference was itself brighter, relative to the pre-storm reference, in more vulnerable tracts (coefficient 7.7, 5.7 to 9.7; average total impact 8.8, 6.7 to 11.1). These estimates reflect the choice of reference nights as much as the outage, and Section 3.4 shows that single-night SVI coefficients of this size are within ordinary variation. We therefore do not read them as differences in recovery between communities.

LISA of 16 February relative radiance found 98 high–high and 145 low–low tracts at $p < 0.05$, and 13 and 79 after false-discovery-rate control (Figure 8c; Table S5). None

remained after Bonferroni correction, a level that 999 permutations cannot reach. Low–low clusters lie mainly in the west and northeast of the county, and the largest high–high cluster lies in the east near Galveston Bay. The bivariate LISA flagged 78 tracts (pre-storm reference) and 87 tracts (March reference) on 16 February after false-discovery-rate control, and none on any ordinary night. This statistic depends strongly on how clustered the outcome is, and 16 February dimming was far more clustered than on ordinary nights. We therefore treat these tracts as places where SVI values and clustered dimming coincide, not as evidence of local vulnerability effects.

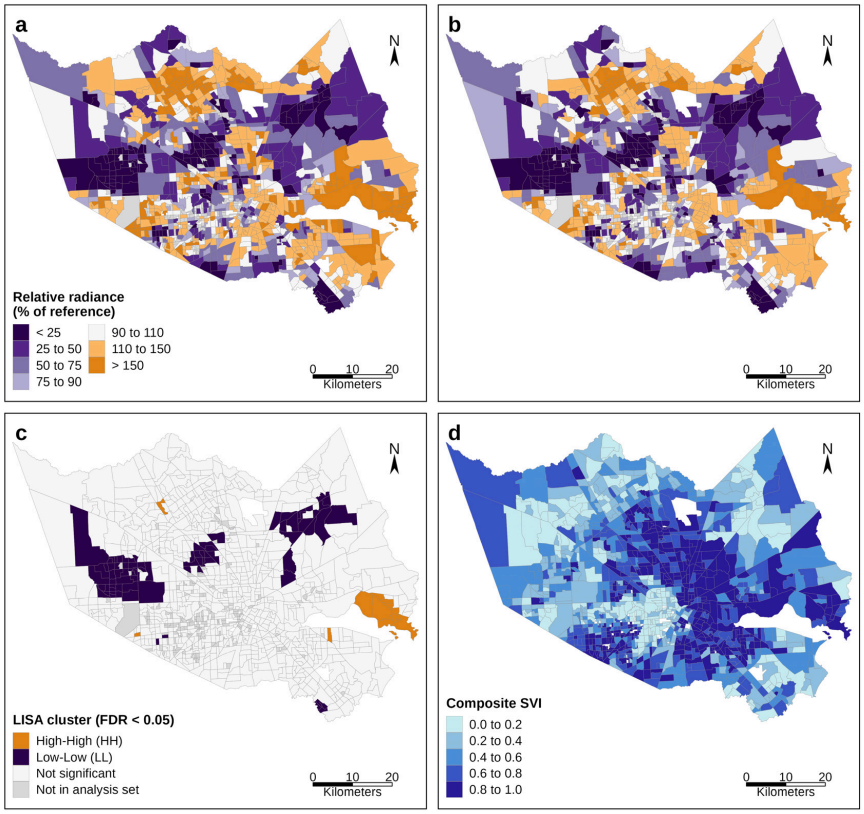


Figure 8. 16 February relative radiance by tract (percent of reference; 100 = as bright as on ordinary clear nights) with (**a**) the pre-storm and (**b**) the March reference; (**c**) LISA clusters of (**a**) after false-discovery-rate control; (**d**) composite SVI. Grey tracts in (**a**–**c**) are outside the analysis set.

4. Discussion

4.1. What the Observations Support

The observations support three conclusions. First, the outage left a clear signal on 16 February. On the common set, 35% to 37% of pixels were below half their reference, against at most 6% on ordinary clear nights, and the dimming was strongly clustered (Moran's I of 0.57 and 0.58, against at most 0.38). Second, light had returned by the first clear nights after the outage for most pixels that dimmed deeply: 92% of deep declines with an observed return were lit again by 21 February, which fits the county outage record (Figure 3). Third, the observations cannot place the return within the unobserved nights between 16 February and 19 to 21 February. Detectors that smooth the series or require persistence assign dates inside or after this interval, but the controlled experiment shows that these dates reflect the rule and the gaps, not the observations (Table 4).

Two features of the event limit what the 16 February image represents. The outages rotated between circuits [2], so the image shows which areas were dark at about 02:48 local time, not everyone who lost power at some point. The University of Houston survey [3]

found that 91% of residents lost power at some point, which a single image cannot measure. Also, the share of brightened pixels on 16 February was within the ordinary range. Brightening did not increase overall, but snow may still have brightened particular places (Section 5).

The SVI and density results should be read with these limits in mind. The 16 February SVI coefficients were small and imprecise in every specification and lay inside the spread of ordinary-night coefficients on the same pixels and tracts. The data therefore neither show an association between social vulnerability and 16 February dimming nor rule out sizable associations in either direction: with the pre-storm reference, the average total impact of SVI was 21.4 percentage points of reference radiance, with a 95% interval of −22.7 to 67.8. Density is less clear. In the snapshot set, density coefficients were negative with every weight matrix and model, so denser tracts tended to be darker relative to their reference, but the intervals excluded zero only in the spatial error models and with the broadest land cover mask. In the common set, the 16 February density coefficients were positive in the spatial lag models and negative in the spatial error models, and all lay within the ordinary-night range. We read this as a weak, model-dependent pattern, not an established difference. Finally, with the detector-based workflow, more vulnerable tracts appeared to recover faster. That result rested on detector dates, a two-day smoothing window and tract averages that depended strongly on which pixels contributed, and its sign reversed under a five-day persistence rule.

4.2. Relation to Previous Studies

Like Xu et al. [16], we found 16 February to be the clearest night of the outage; in Harris County it is the only outage night with flag-0 observations. They also found that NTL showed more outage in Harris County than the county tracking data did. We see the same pattern: 35% of pixels were below half their reference at the overpass, while EAGLE-I recorded about 15% of modeled customers out. The two sources count different things: EAGLE-I counts customer accounts across the county, whereas our measure counts dark pixels and so weights places by their area of developed land. EAGLE-I also covered only part of the customers in 2021 [32]. The tract-level correlation of Xu et al. with the Hispanic population share ($r = 0.083$) also comes from a single night. Our placebo comparison shows that single-night estimates of community differences vary widely from one ordinary night to another, so such estimates should be compared with that background before they are interpreted. Finally, the quality-flag definitions in Xu et al. [16] follow Collection 1, in which flag 1 marks ephemeral lights; in Collection 2, flag 1 marks poor-quality retrievals [15]. Both studies exclude flag 1.

Zheng et al. [22] found that aggregating pixels to tracts and counties hides much of the variation in hurricane recovery. Our split-half diagnostic points the same way: tract means of pixel recovery days depended strongly on which pixels contributed, and two random halves of the same tract agreed at only $r = 0.21$. Their approach suits recoveries that last months, where gaps of a few days matter little and detected change can be tested against pre-event variation. The outage caused by Uri lasted days, and the gaps around it were as long as the outage, so the same kind of data can bound recovery only to the spacing of clear nights.

Best et al. [10] show what utility records make possible: times to 50%, 80% and 95% restoration from ZIP code records at 15 min intervals, analyzed with spatial Durbin models and reported as direct, indirect and total effects. We follow their reporting of spatial impacts. For a short outage, recovery timing at tract level needs records of that kind; NTL is better used to show where dimming occurred on the nights that were observed.

4.3. Recommendations for Reporting

Several practices would make NTL outage and recovery studies easier to check and compare. Studies should state the product collection, band and quality-flag definitions, because flag meanings differ between collections. They should give image dates as local overpass dates, state where date windows end and report overpass times. They should report the share of usable observations for each night. Recovery should be assessed only after an observed decline, with pixels that showed no decline or had no observation reported separately. It should be reported as bounds between observed nights, with unresolved cases counted, rather than as smoothed detector dates. Detection rules should be tested against known recovery dates under the real observation masks. Finally, event-night estimates, including estimates of community differences, should be compared with the same estimates on ordinary nights for the same pixels and areas, and reported with confidence intervals and spatial impacts.

4.4. Explanations and Hypotheses

Some patterns fit how the outage was managed, but our data cannot test these explanations. Rotating outages are carried out by switching distribution circuits off and on [2], so pixels served by the same circuit would be dark or lit together at the time of the overpass. This fits the strong clustering of 16 February dimming and the low–low clusters in the west and northeast of the county (Figure 8). The number of circuits with critical load limited how far utilities could rotate outages [2], and customers on feeders shared with emergency services tend to be restored sooner [10]. Together with commercial and industrial lighting, this could explain why more developed tracts tended to dim less (Section 3.5), although that estimate was imprecise with the pre-storm reference. The high–high cluster near Galveston Bay could reflect industrial facilities or local snow brightening. Testing these explanations would need circuit-level load-shed schedules, the locations of exempt circuits and critical facilities, and the utility's restoration order. Without such data they remain hypotheses.

5. Limitations of the Study

Several limits remain.

Temporal coverage. With flag-0 observations only, 16 February is the only usable night during the outage, and the next are 19 to 21 February. Recovery can be bounded only by these nights. When light returned within the gap is not observed, and outages on other nights, including the rotating outages of 15 and 17 February, are not seen.

What light measures. Radiance measures emitted light, not electricity supply. Pixels that stayed bright may contain circuits that were not shed, such as those serving critical facilities, or lighting on backup generation. A return of light can also reflect people returning home, or be delayed by building damage. With the data used here, these cannot be separated from restored service. A pixel with no observed decline is therefore not evidence of uninterrupted service, and a return of light is not a restoration date.

Snow and residual observation effects. MODIS observes in daylight and cannot detect snow present only at night, and 226 of the 922 snapshot tracts lacked adequate MODIS observation. Flag-1 retrievals on 15 and 18 February may include outliers caused by the outage itself, which cannot be separated from cloud or other contamination. Viewing-angle effects remaining after the product's correction were not modeled; the placebo comparison bounds their combined influence only to the extent that ordinary nights span similar conditions.

Reference sensitivity. Relative radiance depends on the reference nights. The March reference was brighter than the pre-storm reference in more vulnerable tracts, so estimates for the nights after the outage change with the reference and are not interpreted as differences in recovery.

Aggregation. When pixels within a tract vary widely, a tract mean depends on which pixels contribute, as the split-half diagnostic for recovery days shows; part of this variation may be real differences within tracts. Tract values also depend on the reduction grid, the weighting and the tract polygons used to assign pixels (Table 5).

Scale of outage validation. EAGLE-I reports county-level customer counts compiled from utility reports and covered roughly 82% to 94% of Texas customers in 2021 [32]. It can check the event timeline but cannot validate tract or pixel estimates, and no tract-level outage records were available.

Scope. The analysis covers one event, one county and overpasses between 01:26 and 02:54 local time, with SVI from 2016 to 2020 survey data. Events with longer outages or more clear nights may allow more precise timing.

6. Conclusions

Daily nighttime lights showed clearly where Harris County was dark early on 16 February 2021, and that most of those areas were lit again by 19 to 21 February. They could not determine when light returned between the 16 February observation and the next clear observations on 19–21 February. Tract-level recovery durations derived from them depended on smoothing, persistence and aggregation choices, not on the observations. With a detector-based workflow, the association between social vulnerability and recovery time even reversed sign when five instead of three consecutive days were required, a pattern consistent with noisy observations breaking runs more often in some tracts than in others. A single-night estimate of the association between social vulnerability and dimming was imprecise and could not be distinguished from the variation on ordinary clear nights, so the data neither support nor rule out such an association.

For short outages, the spacing of usable nights sets the precision with which satellite data can date recovery. Studies that use NTL to compare outage or recovery between communities should report observation availability, require an observed decline, report censoring and compare event-night estimates with ordinary nights. Tract-level recovery timing for events like Uri will need utility or customer records at a finer scale than those now publicly available.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/ijgi15110491/s1, Table S1: Observation coverage, clear-night variability and the detector-based workflow by SVI quartile (tract means); Table S2: Snow status on 15 February and 16 (MODIS Terra and Aqua) for the 922 tracts of the main snapshot, and models with snow; Table S3: 16 February relative radiance (percent of reference) modeled with alternative spatial weights and models (922 tracts); Table S4: 16 February relative radiance (percent of reference) with alternative land cover masks (spatial lag models, queen weights); Table S5: LISA clusters of 16 February relative radiance (pre-storm reference; 922 tracts; 999 permutations); File S1: Google Earth Engine and R scripts, Earth Engine exports and tract inputs used in this study.

Author Contributions: Conceptualization, Md Jubier Ahammed and Chao Xu; methodology, Md Jubier Ahammed, Chao Xu and Yaping Xu; software, Md Jubier Ahammed; validation, Md Jubier Ahammed, Chao Xu, Yaping Xu and Hanlin Zhou; formal analysis, Md Jubier Ahammed; investigation, Md Jubier Ahammed; resources, Md Jubier Ahammed and Chao Xu; data curation, Md Jubier Ahammed; writing—original draft preparation, Md Jubier Ahammed; writing—review and editing, Md Jubier Ahammed, Chao Xu, Yaping Xu and Hanlin Zhou; visualization, Md Jubier Ahammed and Chao Xu; supervision, Chao Xu; project administration, Chao Xu. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: NASA Black Marble VNP46A2 and VNP46A1 data are available from NASA LAADS DAAC and Google Earth Engine; MODIS snow products from the NSIDC DAAC;

EAGLE-I outage records from figshare (https://doi.org/10.6084/m9.figshare.24237376); CDC/ATSDR SVI 2020 from the CDC/ATSDR website, https://www.atsdr.cdc.gov/place-health/php/svi/svi-data-documentation-download.html (accessed on 26 October 2026); and NLCD 2019 from the USGS (https://www.mrlc.gov). The Google Earth Engine and R scripts, the Earth Engine exports, the Harris County EAGLE-I subset and the tract inputs used in this study are provided in the Supplementary Materials (File S1).

Acknowledgments: During the revision of this manuscript, M.J.A. used Claude (Anthropic; Claude Opus 5.5, accessed October 2026) to assist with drafting analysis and figure code, language editing and checking reported numbers against the analysis outputs. C.X. used ChatGPT (OpenAI; GPT-6, accessed October 2026) to assist with manuscript and code review and with developing Python scripts for independent recalculation from the supplied data. C.X. also used ChatGPT's image-generation feature (OpenAI, accessed October 2026) to redesign Figure 2 from the authors' original workflow diagram, including its layout, colors, icons and connectors. The authors reviewed and edited all AI-assisted output, verified the code as described in Section 2.9, and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

ACS	American Community Survey
AIC	Akaike Information Criterion
ATSDR	Agency for Toxic Substances and Disease Registry
CDC	Centers for Disease Control and Prevention
CI	Confidence interval
DAAC	Distributed Active Archive Center
DTR	Days to recovery
EAGLE-I	Environment for Analysis of Geo-Located Energy Information
ERCOT	Electric Reliability Council of Texas
FDR	False discovery rate
LAADS	Level-1 and Atmosphere Archive and Distribution System
LISA	Local Indicators of Spatial Association
MODIS	Moderate Resolution Imaging Spectroradiometer
NDSI	Normalized Difference Snow Index
NLCD	National Land Cover Database
NSIDC	National Snow and Ice Data Center
NTL	Nighttime light
SD	Standard deviation
SVI	Social Vulnerability Index
TIGER	Topologically Integrated Geographic Encoding and Referencing
USGS	United States Geological Survey
VIIRS	Visible Infrared Imaging Radiometer Suite

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